VIVARIUM ORIGINS-OF-LIFE SIMULATION CAMPAIGN

Building from serpentinization-driven alkaline hydrothermal vents to protometabolism, protocells, heredity, and autonomous life

Core modeling idea: Treat the origin of life as a sequence of coupled, independently falsifiable transitions: geochemical disequilibrium -> carbon fixation -> concentration and compartmentalization -> protometabolic networks -> protocells -> hereditary chemistry -> autonomous cells.

Prepared as a theoretical, empirical, and simulation-design foundation for a modular Vivarium campaign.

Scope: deep-sea serpentinization-driven alkaline vent hypothesis, with explicit comparison to shallow-vent, terrestrial geothermal, and cyanosulfidic alternatives. Literature verified through October 2026.

Executive summary

The strongest version of the alkaline hydrothermal-vent hypothesis is not “life began at a vent” as a single claim. It is a chain of mechanistic propositions linking water-rock reactions to biological organization. That makes it unusually suitable for Vivarium: each proposition can be represented as a process with explicit state variables, rate laws, spatial interfaces, uncertainty distributions, and alternative implementations.

The proposed campaign therefore begins below biology. A geochemical world supplies H2, alkalinity, heat, reduced minerals, and sustained flow through serpentinization. Vent fluids encounter CO2-bearing ocean water across dynamically precipitating mineral interfaces. That interface can generate pH and redox disequilibria, catalyze carbon reduction, and create pores that concentrate reaction products. The campaign then asks whether experimentally supported reaction networks can become persistent, self-reinforcing, compartmentalized, and eventually coupled to membranes and heredity.

Crucially, the model should not hard-code the alkaline-vent narrative as the answer. Recent work strengthens several links - including H2-dependent mineral-catalyzed CO2 fixation, pH-gradient-driven formate formation, mixed-amphiphile vesicle formation, and nonenzymatic metabolic chemistry - while other links remain uncertain. A major 2026 critique argues that in-situ vent pH gradients, sulfide availability, vent longevity, and hydrocarbon kinetics may be substantially less favorable than classical alkaline-vent models assume. The campaign should represent those objections as parameterizations and competing process models, so that the simulations adjudicate among scenarios rather than illustrate one preferred story. [28]

Primary scientific question Can a physically plausible early-Earth hydrothermal system generate and sustain localized, increasingly autonomous chemical organizations - without inserting metabolism, membranes, or heredity by assumption?

Report map

- 1. Scientific framing and hypothesis graph
- 2. Mechanistic modules and evidence
- 3. Quantitative parameter and data provenance
- 4. Vivarium process-bigraph architecture
- 5. Staged simulation campaign
- 6. Discriminating experiments and falsification tests
- 7. Competing origin environments
- 8. Data acquisition priorities
- 9. Evidence gaps and risk register
- 10. Annotated reference library

1. Scientific framing: from geology to biology

The campaign should represent origins as a succession of nonequilibrium transitions. A useful state-space view is:

Stage	Dominant organization	Key emergent quantity	Simulation question
0	Ultramafic crust + water	H2, alkalinity, heat	Does serpentinization provide sustained reducing power at plausible rates?
1	Vent-ocean flow reactor	pH/redox/thermal gradients	Are useful gradients maintained at relevant in-situ T, P, salinity and geometry?
2	Mineral microcompartments	Catalytic surfaces + concentration	Do pores retain and concentrate products faster than dilution/destruction removes them?
3	Protometabolic network	Persistent carbon flux	Do reaction subnetworks become self-reinforcing under continuous feed?
4	Hybrid mineral-organic compartments	Selective permeability	Can amphiphiles stabilize compartments without destroying access to environmental gradients?
5	Protocells	Internalized free-energy conversion	Can selection favor systems progressively less dependent on vent geometry?
6	Heritable chemical systems	Memory + variation	Can polymers or catalytic ensembles preserve and propagate functional organization?
7	Pre-LUCA ecology	Autonomous metabolism + heredity	What innovations permit escape from the vent and diversification?

1.1 Why alkaline, serpentinization-driven vents?

The model should distinguish these systems from high-temperature black smokers. Lost City-type systems are driven primarily by exothermic water-rock reactions in ultramafic rocks rather than by a shallow magma chamber. The discovery paper reported ~40-75 C vent fluids at pH 9.0-9.8; later work documented fluids up to ~91 C and pH as high as ~10.8, carbonate chimneys, and long-lived hydrothermal activity. [5] [6] [7] [8]

The theoretical attraction is that serpentinization supplies a continuous physical source of chemical disequilibrium: reduced H2-rich alkaline fluids encounter relatively oxidized carbon-bearing ocean water. Russell, Martin, Lane, Sojo and colleagues developed the idea that mineral

microcompartments and Fe/Ni-bearing precipitates could act as primitive electrochemical reactors whose topology foreshadows membrane bioenergetics. [1] [2] [3] [4]

1.2 The hypothesis graph to encode

Link	Mechanistic proposition	Evidence class	Current status
A	Serpentinization generates H2, alkalinity, heat and reduced mineral phases.	Field + geochemical theory	Strong
B	Flow/mixing builds persistent pH, redox and thermal gradients across pores or precipitates.	Field analog + reactor studies + theory	Moderate; challenged in 2026
C	Fe/Ni/S/oxide minerals catalyze H2-driven CO2 reduction to biologically relevant C1-C3 compounds.	Laboratory experiments	Strong for selected conditions
D	Continuous flow plus surfaces and thermophoresis overcome dilution.	Transport models + laboratory analogs	Moderate-strong
E	Nonenzymatic chemistry organizes into metabolic-like reaction networks.	Laboratory reaction networks	Moderate; environmental matching incomplete
F	Amphiphile mixtures form stable, selectively permeable compartments under vent-like conditions.	Laboratory experiments	Moderate
G	Environmental proton gradients can be progressively internalized as membrane bioenergetics.	Comparative biochemistry + models	Plausible but indirect
H	Nucleotide/polymer chemistry can arise and become heritable in this setting.	Partial analog experiments	Weakest link
I	The trajectory converges toward pre-LUCA/LUCA physiology.	Phylogenomic reconstruction	Useful boundary condition, not direct origin evidence

2. Mechanistic modules and evidence base

2.1 Module A - serpentinization and geochemical energy supply

Mechanism. Water circulating through Fe(II)-bearing ultramafic rocks hydrates olivine and related minerals, producing serpentine/brucite assemblages and oxidizing part of the iron toward Fe(III)-bearing phases such as magnetite. The associated electron transfer can reduce water to H2. The resulting fluids can become strongly alkaline and reducing; H2 then serves as an electron

donor for subsequent carbon reduction. The 2023 synthesis by Schwander et al. is the best current gateway because it integrates energy, electrons, organics, catalysts, nutrients, gradients, modern analogs and astrobiological context. [9]

Provenance. Lost City field observations constrain fluid temperature, pH, mineralogy, carbon species and system longevity. Kelley et al. established the off-axis serpentinite-hosted system; Fruh-Green et al. documented at least 30,000 years of activity; Ludwig et al. linked carbonate chimney evolution to serpentinization-controlled fluid chemistry. [5] [7] [8]

Vivarium process serpentinization(rock_state, water_flux, T, P) -> H2_flux, alkalinity_flux, heat_flux, mineral_transformations, dissolved_metals. Treat the output as a distribution, not a fixed H2 concentration.

2.2 Module B - hydrothermal flow, mixing, precipitation and pores

The vent-ocean interface is a coupled reactive-transport problem. Alkaline vent fluid mixes with carbon-bearing ocean water; temperature, ionic strength and redox state change; carbonate, silicate, sulfide and oxide phases may precipitate; permeability evolves; and pore geometry feeds back on flow. Herschy et al. built a bench-top origin-of-life reactor specifically to reproduce continuous alkaline fluid injection into a more acidic ocean analog and study precipitating barriers. [10]

A key modeling choice is whether thin Fe(Ni)S-rich barriers are abundant and persistent under Hadean conditions. Classical formulations assume such barriers can sustain steep gradients; Tutolo 2026 argues that early ocean sulfur scarcity, in-situ pH behavior and serpentinizing-system geology may make that configuration far less plausible. The campaign should implement both versions. [4] [28]

2.3 Module C - natural electrochemistry and pH-gradient-driven carbon reduction

Hudson et al. provided a particularly important mechanistic constraint: in a microfluidic system, CO_2 was reduced with H_2 to formate at room temperature and 1.5 bar when acidic and alkaline streams were separated across freshly precipitated Fe(Ni)S material. Isotopic labeling supported carbon provenance, and deuterium experiments favored an electrochemical-cell interpretation with spatially separated half-reactions rather than simple direct hydrogenation. Reducing the gradient, removing H_2, or removing the precipitate eliminated detectable product. [11]

Altair et al. 2025 extended this logic in a macroscale reactor using mineral preparations intended to mimic chimney formation. They detected formic and acetic acid production and examined Ni incorporation, temperature, passive versus induced currents, and current magnitudes from roughly nanoamps to microamps. This is an unusually direct source of parameterization for an electrochemical Vivarium process. [12]

Key discriminating variable Do gradients persist at in-situ temperature and pressure, or are

large pH differences primarily an artifact of cooling/measurement? Encode pH(T,P,ionic_strength), not laboratory pH as a constant environmental state.

2.4 Module D - mineral-catalyzed carbon fixation

The experimental bridge from geochemistry toward metabolism is now one of the strongest parts of the hypothesis. Huber and Wachtershauser demonstrated formation of activated acetate/thioester chemistry on coprecipitated FeS/NiS under hydrothermal conditions. Varma et al. later showed native Fe can reduce CO2 toward intermediates and products of the acetyl-CoA pathway. Preiner et al. showed that greigite, magnetite and awaruite catalyze H2-dependent CO2 fixation at 100 C under alkaline aqueous conditions, producing formate, acetate, pyruvate, methanol and methane. [13] [14] [15]

The Preiner study is especially useful quantitatively: formate reached up to ~200 mM, acetate up to ~100 uM, pyruvate up to ~10 uM, and methanol up to ~100 uM under tested conditions. These should not be treated as environmental concentrations; they are laboratory outcomes that constrain possible reaction yields/rates under specific mineral, temperature and pressure regimes. [15]

2.5 Module E - protometabolic network formation

Rather than inserting a modern metabolic pathway, the campaign should begin with an experimentally motivated reaction graph and ask whether metabolism-like subnetworks emerge under sustained flow. Muchowska et al. showed that metals can promote six of eleven rTCA reactions, including two three-step sequences, and observed reductive amination to amino acids. A later Fe2+-promoted network generated nine of eleven TCA intermediates and all five universal metabolic precursors from pyruvate and glyoxylate, with related amino-acid chemistry. [16] [17]

This evidence supports a network-level modeling strategy: elementary or lumped reactions carry rate laws, mineral dependencies and environmental windows. Persistent flux cycles, autocatalytic motifs, cross-catalysis and product retention become emergent observables rather than assumptions.

2.6 Module F - transport, concentration and retention

Any oceanic origin scenario must defeat dilution. Baaske et al. modeled thermally driven convection plus thermophoresis in elongated hydrothermal pores and predicted extreme nucleotide accumulation, exceeding 10^8-fold in favorable modeled geometries. Mast and Braun later demonstrated a thermal trap that simultaneously supported DNA replication cycling and thermophoretic accumulation in a laboratory temperature gradient. [18] [19]

For the campaign, concentration should arise from the balance of advection, molecular diffusion, thermodiffusion, adsorption/desorption, precipitation, degradation and pore geometry. This is a natural place to connect a finite-volume spatial transport process to chemical network processes.

2.7 Module G - mineral compartments to organic protocells

Mineral pores can provide initial spatial individuality without requiring lipid membranes to exist first. The later transition to organic boundaries is experimentally plausible but condition-dependent. Jordan et al. tested heterogeneous mixtures of single-chain amphiphiles and found vesicle formation across a broader range of alkaline pH, salinity, divalent cations and elevated temperatures than simple single-fatty-acid models imply. [20]

The simulation should model amphiphile production/import, partitioning, adsorption to minerals, vesicle nucleation, permeability, growth, division and leakage. A central evolutionary tension is that early membranes must provide enough retention to support selection while remaining permeable enough to exploit external geochemical gradients.

2.8 Module H - bioenergetic internalization

Lane and Martin proposed that natural proton gradients across inorganic barriers could precede membrane bioenergetics, with initially leaky protocell membranes later becoming less permeable and forcing the evolution of active ion pumping. Sojo et al. developed a quantitative model in which LUCA-like cells can use proton gradients only if membranes remain sufficiently leaky, linking gradient dependence to later bacterial/archaeal membrane divergence. [3] [21]

In Vivarium, this becomes an evolutionary control problem: vary membrane permeability, Na^+/H^+ coupling, environmental gradient magnitude and catalyst efficiency; reward carbon/energy throughput and compartment persistence; then determine whether selection favors increasingly internalized gradient generation.

2.9 Module I - nitrogen, phosphorus and other limiting resources

Carbon receives most attention, but origins may be controlled by less abundant resources. Dorr et al. reported NH3 formation from N2 using freshly precipitated FeS and H2S at ~70-80 C; the reported yield was of order 0.1% relative to FeS, emphasizing that nitrogen fixation may be possible yet flux-limiting. [22]

Phosphorus is another likely bottleneck. Schreibersite, (Fe,Ni)3P, provides a reduced and reactive P source capable of generating activated phosphorus species; Pasek and colleagues have emphasized the importance of mineral-organic P chemistry and environmental pH. The 2017 schreibersite analysis notes that acidic conditions favor activated phosphate formation whereas basic conditions can favor organophosphates, a useful example of environmental incompatibility that may require transport between niches. [23]

2.10 Module J - polymers, information and heredity

This is the least complete link in a strict deep-sea alkaline-vent scenario. Hydrothermal pores can plausibly concentrate nucleotides and polymers, and Fe/Ni/S chemistry can promote peptide formation, but no single experimentally validated vent pathway currently spans feedstock synthesis, nucleotide activation, template copying, functional polymers, compartment coupling and open-ended heredity. Huber and Wachtershauser demonstrated peptide formation from

amino acids using CO and (Ni,Fe)S at 100 C and pH 7-10, establishing one possible route to early peptides. [24]

The campaign should therefore treat informational chemistry as a branch point, not a scripted milestone. Candidate modules can include RNA-world chemistry, peptide-first catalysis, compositional heredity, mineral-surface templating, and exchange with shallow or terrestrial environments.

2.11 LUCA as a far-end boundary condition

LUCA is not the first life, but phylogenomic reconstructions provide a distant target state. Weiss et al. reconstructed 355 protein families with evidence consistent with an anaerobic, H2-dependent, CO2-fixing, FeS-rich physiology associated with hydrothermal settings. Moody et al. 2024 inferred a much larger and earlier LUCA (~4.2 Ga in their analysis) embedded in an ecosystem, highlighting that the transition from origins chemistry to LUCA likely contains substantial prior evolution. [25] [26]

Use LUCA reconstructions as constraints on which capabilities must eventually be acquired, not as evidence that the first replicators had LUCA-like complexity.

3. Quantitative parameter and data provenance

The following table separates observations from laboratory analogs and theoretical values. Every parameter should carry source metadata, experimental context, uncertainty and a provenance class inside the model configuration.

Quantity	Representative value/range	Provenance	How to use
Lost City vent T	~40-75 C discovery; later measurements to ~91 C	Field observation [5] [8]	Prior over outlet conditions; do not equate outlet T with reaction-zone T.
Lost City pH	~9.0-9.8 discovery; later ~9-10.8	Field/lab-reported fluid chemistry [5] [8]	Use with temperature/pressure correction; include 2026 critique.
System longevity	>=30,000 y directly documented; potentially much longer	Isotopic dating [7]	Sets duration of persistent forcing; explore shorter ancient systems.
CO2 + H2 -> formate	Room T, ~1.5 bar in microfluidic gradient experiment	Laboratory [11]	Fit reaction probability/rate to gradient, catalyst and gas availability.
Mineral H2/CO2 products	100 C; formate up to ~200 mM; acetate ~100 uM; pyruvate ~10 uM	Laboratory [15]	Calibration targets for mineral-catalysis module, not environmental concentrations.
Thermophoretic enrichment	>10^8-fold in favorable modeled pore geometries	Transport model [18]	Use to validate transport implementation across pore geometries.
Thermal trapping	DNA accumulation time ~92 s; doubling ~50 s in PCR analog	Laboratory analog [19]	Demonstrates coupled thermal cycling + retention; not a prebiotic replication rate.

Quantity	Representative value/range	Provenance	How to use
FeS N2 -> NH3	~70-80 C; ~0.1% yield relative to FeS	Laboratory [22]	Potential fixed-N source with large uncertainty and surface-age dependence.
Amphiphile vesicles	Formation under alkaline/saline/divalent-cation conditions tested	Laboratory [20]	Defines feasible region for membrane-assembly state transitions.
Electrochemical current	~10 nA to 10 uA sufficient in 2025 reactor experiments	Laboratory [12]	Scale current-density/electron-transfer module to geometry and surface area.

3.1 Provenance schema for simulation parameters

- FIELD: measured in a natural hydrothermal or serpentinizing system. Preserve site, depth, temperature/pressure convention, sample handling and analytical method.
- LAB-DIRECT: measured in an experiment designed to reproduce a proposed mechanism. Preserve reactor geometry, feed chemistry, catalyst preparation, pressure, temperature, duration and detection limit.
- LAB-ANALOG: demonstrates a physical principle relevant to vents but uses biological or otherwise non-prebiotic components (for example PCR in a thermal trap).
- MODEL: derived from transport, thermodynamic, kinetic, phylogenetic or evolutionary modeling. Preserve equations, assumptions and priors.
- RECONSTRUCTION: inferred ancestral physiology/genome state. Treat as a probabilistic boundary condition.
- HYPOTHESIS: mechanistic proposition not yet independently constrained. This category should never silently become a fixed parameter.

3.2 Data object design

Each quantitative object in the campaign should be accompanied by a machine-readable provenance record. A minimal record is: source DOI; evidence class; system/site; measurement conditions; variable definition and units; central value; uncertainty or range; transformation applied; original table/figure; model module consuming the value; and whether the value is calibration, validation or prior information.

4. Proposed Vivarium process-bigraph architecture

A process-bigraph implementation is well matched to the scientific problem because the relevant objects are nested compartments and interfaces with processes operating on incompatible timescales. The architecture should allow models to be exchanged without changing the whole campaign.

Process	State / ports	Possible implementation	Natural timescale
Geochemical rock reactor	rock minerals, water, T, P ->	kinetic/thermodynamic	days-years+

Process	State / ports	Possible implementation	Natural timescale
	H2, alkalinity, ions, heat	serpentinization model	
Hydrothermal circulation	pressure, permeability, heat -> flow field	Darcy/advection model; reduced network initially	s-days
Reactive precipitation	ion activities -> mineral phases, porosity, conductivity	kinetic mineral precipitation	s-days
Electrochemistry	pH/Eh, catalysts, H2, CO2 -> electron/current + products	Nernst + empirical kinetics from reactor data	ms-hours
Spatial transport	species fields, T field, flow -> advection/diffusion/thermophoresis	finite volume spatial-transport	ms-hours
Surface chemistry	adsorbed species + minerals -> organics	reaction-network kinetics	s-days
Protometabolism	organic network -> fluxes/cycles	ODE/stochastic reaction network	s-days
Amphiphile dynamics	lipids + surfaces -> vesicles/permeability	coarse-grained state model	s-hours
Compartment ecology	pores/protocells -> growth, fission, extinction	agent-based population process	min-generations
Polymer/heredity	monomers/polymers -> copying/catalysis/variation	stochastic polymer model	s-generations
Evolution	heritable traits -> selection and innovation	population genetics / adaptive search	generations+

4.1 Core shared stores

- geometry: pore network, surfaces, membrane compartments, interface areas
- fluid fields: velocity, pressure, temperature, pH, ionic strength, redox potential
- chemical species: dissolved inorganic carbon, H2, sulfur species, nitrogen species, phosphorus species, metals, organics
- solid phases: olivine/serpentine/brucite/magnetite, Fe/Ni sulfides, carbonates, phosphides
- electrical state: potentials, conductivity, current density
- organic organization: amphiphiles, polymers, catalysts, reaction-network state
- lineage state: compartment ancestry, inherited catalysts/polymers, fitness observables
- provenance: parameter/source IDs and uncertainty ensemble membership

4.2 Multiscale coupling principle

Do not force every process onto one timestep. Fast chemical/electrochemical processes can be stepped locally or solved to quasi-steady state inside slower transport updates; mineral evolution and geological forcing can run on much slower clocks; evolutionary experiments can coarse-grain many chemical episodes into one reproductive event. Vivarium should make the coupling explicit rather than burying it in one monolithic simulator.

5. Staged simulation campaign

Campaign 0 - reproduce the physical analog

Build a Lost City-like baseline and verify fluid mixing, heat transport, mineral precipitation and H2/alkalinity delivery against published field constraints. No origin-of-life claim yet. Output: a validated geochemical/spatial testbed.

Campaign I - geology -> reduced carbon

Add H2-driven CO2 reduction on mineral surfaces and across gradients. Reproduce Hudson, Preiner and Altair experimental regimes before extrapolating. Ask which combinations of T, pH, pressure, H2, CO2 and mineralogy generate sustained net organic-carbon flux.

Campaign II - reduced carbon -> protometabolism

Insert broad reaction graphs informed by Huber/Wachtershauser and Muchowska/Moran chemistry. Search for persistent cycles, flux amplification, product reuse and autocatalytic motifs under continuous feed and washout.

Campaign III - dilution -> localized chemical individuals

Add explicit pore geometry, adsorption, convection and thermophoresis. Ask when retention time and concentration cross thresholds for network persistence. Let pore morphology evolve through precipitation/dissolution.

Campaign IV - mineral compartments -> protocells

Introduce mixed amphiphiles and membrane assembly. Evolve permeability, adsorption, vesicle growth and fission. Measure whether organic boundaries enhance persistence without cutting systems off from geochemical energy.

Campaign V - environmental -> internal bioenergetics

Allow catalytic ion transport and later active pumping. Test whether selection can move from externally imposed gradients toward internally maintained electrochemical states.

Campaign VI - chemistry -> heredity

Compare RNA-like, peptide-assisted, compositional and hybrid heredity modules. Require inherited state to improve persistence/throughput, not merely template replication in isolation.

Campaign VII - ecology before LUCA

Run heterogeneous populations in connected pore networks with cross-feeding, parasitism, cooperation and horizontal exchange. Compare emergent capabilities to ancestral reconstructions without assuming LUCA is primitive.

Campaign VIII - escape from the vent

Relax environmental support. Vary gradient strength, mineral access and feed chemistry. Identify the minimum innovations required for autonomous persistence away from the geological reactor.

5.1 First publication-sized milestone

Recommended first target Can a dynamically precipitating, spatially resolved alkaline vent-ocean interface generate and retain a persistent protometabolic carbon flux under parameter ranges consistent with field observations and laboratory experiments?

This target is intentionally pre-genetic. It is difficult enough to be scientifically meaningful, admits direct laboratory validation, and avoids hiding the weakest part of the origin story behind assumed replicators.

6. Discriminating experiments and falsification tests

Test	Vent-model prediction	Competing outcome	Model observable
In-situ pH correction	Large usable gradient persists across reaction interface	Gradient collapses substantially at hydrothermal T/P	delta pH, proton motive contribution, carbon-fixation rate
Sulfur availability	Enough sulfide precipitates catalytic Fe(Ni)S interfaces	Sulfide-poor system dominated by carbonates/oxides	mineral area fractions; conductivity; catalytic yield
Pressure/H2 dependence	CO2 reduction remains effective across plausible subseafloor regimes	Products require laboratory-favorable H2 pressures	rate surface over H2 fugacity and pressure
Pore lifetime	Reactive compartments persist long enough for enrichment/network growth	Precipitation/dissolution/flow resets pores too fast	residence-time distribution; lineage persistence
Dilution	Transport creates concentration hotspots	Washout dominates chemistry	Damkohler/Peclet/Soret-dependent enrichment
Reaction-network closure	Sustained feed yields self-reinforcing network motifs	Products remain disconnected one-off syntheses	cycle flux, autocatalytic gain, network persistence
Membrane transition	Mixed amphiphiles increase individuality and selection	Salts/T/pH destroy or over-seal membranes	vesicle survival, permeability, resource flux
Fixed N/P supply	N/P production/import supports increasing complexity	N or P becomes hard bottleneck	limiting-resource shadow price / growth sensitivity
Autonomy	Evolution internalizes energy conversion	Systems remain obligately tied to exact vent interface	survival as external gradient is reduced

6.1 What would seriously weaken the deep-sea vent program?

- If realistic in-situ temperature/pressure/ionic corrections eliminate the electrochemical gradients needed for proposed carbon reduction across most plausible geometries.

- If Hadean sulfur inventories make Fe(Ni)S catalytic barriers rare while alternative carbonate/oxide mineralogies fail to support comparable chemistry.
- If experimentally constrained reaction rates are too slow relative to pore turnover, dilution and degradation to produce meaningful local accumulation.
- If resource bottlenecks in nitrogen or phosphorus cannot be overcome by flux from the vent system or connected environments.
- If no plausible pathway couples protometabolic chemistry to heritable compartments without importing chemistry that requires a fundamentally different environment.

7. Competing and hybrid origin environments

The right campaign is not “vent versus everything else.” Many successful prebiotic reactions require conditions that cannot coexist in one location. Vivarium can explicitly represent connected environments and material exchange.

Scenario	Strengths	Weaknesses / tests	Reference anchors
Deep alkaline vent	Continuous H2; gradients; minerals; pores; long-lived forcing	Dilution; P/N; nucleotide chemistry; sulfur/pH critique	Russell/Hall; Martin/Russell; Sojo; Tutolo [1] [2] [4] [28]
Shallow alkaline vent	Vent chemistry plus light, tidal/wet-dry and salinity variability	Geological prevalence on Hadean Earth uncertain	Barge & Price [27]
Terrestrial geothermal field	K-rich/low-Na pools; evaporation and wet-dry cycling	Requires subaerial land and transport; different energy regime	Mulkidjanian et al. [29]
Cyanosulfidic / photochemical chemistry	Powerful routes to building blocks under UV-driven chemistry	Requires compatible feedstocks and environmental sequence	Sutherland [30]
Network of environments	Can combine incompatible chemistries by transport/cycling	Adds geological/logistical constraints; easy to overfit	Simulation should penalize transport complexity

8. Data acquisition priorities

8.1 Field datasets

- Lost City: vent-fluid T, pH with measurement convention, H2, CH4, DIC, major ions, trace metals, mineralogy, chimney structure/porosity and time series where available.
- Other serpentinizing systems: Rainbow, Von Damm, Prony, Strytan and continental serpentinite springs to quantify how variable “alkaline vent” chemistry actually is.
- InterRidge global active vent database for distributions of vent setting, depth, temperature and tectonic context; use as environmental prior rather than a Lost City-only world.
- Ancient-ocean geochemical reconstructions: CO2/DIC, sulfate/sulfide, phosphate, Fe and ionic strength. These are critical because modern seawater cannot simply be substituted for Hadean ocean chemistry.

8.2 Laboratory datasets worth digitizing

- Hudson et al. 2020: product vs pH gradient, catalyst composition, H2, pressure and controls.
- Altair et al. 2025: time courses, currents, Ni content, mineral characterization, temperature effects and passive/induced potentials.
- Preiner et al. 2020: product distributions by mineral, temperature, pressure and reaction time.
- Muchowska et al. 2017/2019: reaction yields and condition windows for each network edge; include off-pathway products.
- Jordan et al. 2019: vesicle formation/stability across amphiphile mixtures, pH, salts, divalent ions and temperature.
- Baaske 2007 / Mast & Braun 2010: transport equations, Soret coefficients/geometry assumptions and observed trapping kinetics.
- Dorr et al. 2003 and phosphorus studies: yields, surface-age effects, pH dependence and detection limits.

8.3 Provenance-first implementation

Every digitized dataset should be stored separately from model code and linked to a DOI plus a condition vector. A parameter fitted from one experiment should retain a pointer to the exact source experiment so later model comparisons can distinguish interpolation from extrapolation. This is essential for a campaign likely to combine dozens of heterogeneous papers.

9. Evidence gaps and model risk register

Risk	Why it matters	Treatment in campaign	Priority
Ancient ocean composition poorly constrained	Controls pH, carbon speciation, minerals and nutrients	ensemble priors + sensitivity analysis	Very high
Lost City is not Hadean Earth	Modern biology/oxidation and geology differ	separate field calibration from ancient extrapolation	Very high
In-situ pH/gradient controversy	Directly affects proposed chemiosmotic driving force	implement competing thermodynamic models	Very high
Catalyst preparation dependence	Fresh FeS/NiS can behave unlike aged natural minerals	surface-state variable + aging kinetics	High
Sparse kinetic data	Many studies report endpoint yields	Bayesian ranges; target new experiments	High
Product detection near limits	False positives/contamination possible in origins chemistry	use controls and isotope-label evidence weighting	High
No complete path to heredity	Prevents claim of full abiogenesis	keep late stages modular and hypothesis-comparative	Very high
Scale gap from reactor to geology	Lab geometry may exaggerate gradients and contact	dimensionless scaling and multiple geometries	High
Narrative bias	Easy to select only favorable papers	encode critiques/negative outcomes as validation targets	Very high

10. Annotated reference library

The list below is a curated foundation for the campaign rather than a claim to exhaust the entire origins-of-life literature. References were selected because they supply a mechanism, quantitative constraint, field observation, model architecture, ancestral boundary condition, or substantive competing interpretation.

[1] Russell, M.J. & Hall, A.J. (1997). The emergence of life from iron monosulphide bubbles at a submarine hydrothermal redox and pH front. Journal of the Geological Society 154, 377-402. DOI: 10.1144/gsjgs.154.3.0377. *Simulation relevance: Foundational alkaline-vent model: FeS compartments at a redox/pH interface; use as historical hypothesis source, not quantitative truth.*

[2] Martin, W. & Russell, M.J. (2007). On the origin of biochemistry at an alkaline hydrothermal vent. Philosophical Transactions of the Royal Society B 362, 1887-1925. DOI: 10.1098/rstb.2006.1881. *Simulation relevance: Connects vent geochemistry to the acetyl-CoA pathway, pterins, purine/pyrimidine precursors and ancestral metabolism.*

[3] Lane, N. & Martin, W.F. (2012). The origin of membrane bioenergetics. Cell 151, 1406-1416. DOI: 10.1016/j.cell.2012.11.050. *Simulation relevance: Theory for transition from environmental gradients to ion-coupled membrane bioenergetics.*

[4] Sojo, V., Herschy, B., Whicher, A., Camprubi, E. & Lane, N. (2016). The origin of life in alkaline hydrothermal vents. Astrobiology 16, 181-197. DOI: 10.1089/ast.2015.1406. *Simulation relevance: Integrated modern statement of the electrochemical flow-reactor hypothesis.*

[5] Kelley, D.S. et al. (2001). An off-axis hydrothermal vent field near the Mid-Atlantic Ridge at 30 degrees N. Nature 412, 145-149. DOI: 10.1038/35084000. *Simulation relevance: Discovery of Lost City; direct constraints on temperature, pH, geology and microbial presence.*

[6] Kelley, D.S. et al. (2005). A serpentinite-hosted ecosystem: the Lost City Hydrothermal Field. Science 307, 1428-1434. *Simulation relevance: Key field synthesis of Lost City geochemistry and ecosystem; use for modern analog constraints.*

[7] Fruh-Green, G.L. et al. (2003). 30,000 years of hydrothermal activity at the Lost City vent field. Science 301, 495-498. DOI: 10.1126/science.1085582. *Simulation relevance: Isotopic dating constraining long-lived serpentinization-driven hydrothermal activity.*

[8] Ludwig, K.A. et al. (2006). Formation and evolution of carbonate chimneys at the Lost City Hydrothermal Field. Geochimica et Cosmochimica Acta 70, 3625-3645. DOI: 10.1016/j.gca.2006.04.016. *Simulation relevance: Chimney mineralogy/evolution and fluid chemistry; important for pore/mineral modules.*

[9] Schwander, T. et al. (2023). Serpentinization as the source of energy, electrons, organics, catalysts, nutrients and pH gradients for the origin of LUCA and life. Frontiers in Microbiology 14:1257597. DOI: 10.3389/fmicb.2023.1257597. *Simulation relevance: Comprehensive current review linking serpentinization to multiple origin-of-life requirements.*

[10] Herschy, B. et al. (2014). An origin-of-life reactor to simulate alkaline hydrothermal vents. Journal of Molecular Evolution 79, 213-227. DOI: 10.1007/s00239-014-9658-4. *Simulation relevance: Experimental reactor architecture; basis for reproducing interface chemistry and flow.*

[11] Hudson, R. et al. (2020). CO2 reduction driven by a pH gradient. PNAS 117, 22873-22879. DOI: 10.1073/pnas.2002659117. *Simulation relevance: Direct experimental evidence for pH-gradient-driven H2/CO2-to-formate chemistry across Fe(Ni)S precipitates, with isotope controls.*

[12] Altair, T. et al. (2025). Carbon reduction powered by natural electrochemical gradients under submarine hydrothermal vent conditions. Journal of the American Chemical Society 147. DOI: 10.1021/jacs.5c01948. *Simulation relevance: Macroscale reactor: carbon reduction, mineral composition, temperature and electrochemical-current measurements.*

[13] Huber, C. & Wachtershauser, G. (1997). Activated acetic acid by carbon fixation on (Fe,Ni)S under primordial conditions. Science 276, 245-247. DOI: 10.1126/science.276.5310.245. *Simulation relevance: Classic Fe/Ni sulfide thioester/acetate experiment connecting geochemistry to activated carbon chemistry.*

[14] Varma, S.J., Muchowska, K.B., Chatelain, P. & Moran, J. (2018). Native iron reduces CO2 to intermediates and end-products of the acetyl-CoA pathway. Nature Ecology & Evolution 2, 1019-1024. DOI: 10.1038/s41559-018-0542-2. *Simulation relevance: Experimental bridge from CO2 to acetyl-CoA-pathway-related compounds using native metals.*

[15] Preiner, M. et al. (2020). A hydrogen-dependent geochemical analogue of primordial carbon and energy metabolism. Nature Ecology & Evolution 4, 534-542. DOI: 10.1038/s41559-020-1125-6. *Simulation relevance: Quantitative H2-dependent CO2 fixation over greigite, magnetite and awaruite; major calibration source.*

[16] Muchowska, K.B. et al. (2017). Metals promote sequences of the reverse Krebs cycle. Nature Ecology & Evolution 1, 1716-1721. DOI: 10.1038/s41559-017-0311-7. *Simulation relevance: Nonenzymatic rTCA reaction sequences and reductive amination; network-edge provenance.*

[17] Muchowska, K.B., Varma, S.J. & Moran, J. (2019). Synthesis and breakdown of universal metabolic precursors promoted by iron. Nature 569, 104-107. DOI: 10.1038/s41586-019-1151-1. *Simulation relevance: Fe2+-promoted network generating TCA intermediates and universal metabolic precursors.*

[18] Baaske, P. et al. (2007). Extreme accumulation of nucleotides in simulated hydrothermal pore systems. PNAS 104, 9346-9351. DOI: 10.1073/pnas.0609592104. *Simulation relevance: Quantitative convection + thermophoresis model for overcoming dilution in vent pores.*

[19] Mast, C.B. & Braun, D. (2010). Thermal trap for DNA replication. Physical Review Letters 104, 188102. DOI: 10.1103/PhysRevLett.104.188102. *Simulation relevance: Laboratory demonstration of thermal cycling plus thermophoretic accumulation; physical analog, not prebiotic proof.*

[20] Jordan, S.F. et al. (2019). Promotion of protocell self-assembly from mixed amphiphiles at the origin of life. Nature Ecology & Evolution 3, 1705-1714. DOI: 10.1038/s41559-019-1015-y. *Simulation relevance: Empirical feasibility map for vesicle formation under challenging salt/pH/temperature conditions.*

[21] Sojo, V., Pomiankowski, A. & Lane, N. (2014). A bioenergetic basis for membrane divergence in archaea and bacteria. PLOS Biology 12:e1001926. DOI: 10.1371/journal.pbio.1001926. *Simulation relevance: Quantitative evolutionary/bioenergetic model of leaky membranes and early ion gradients.*

[22] Dorr, M. et al. (2003). A possible prebiotic formation of ammonia from dinitrogen on iron sulfide surfaces. Angewandte Chemie International Edition 42, 1540-1543. DOI: 10.1002/anie.200250371. *Simulation relevance: Potential mineral-mediated fixed-nitrogen source; low yield and fresh-surface dependence are important constraints.*

[23] Pasek, M.A. et al. (2017). Schreibersite on the early Earth: scenarios for prebiotic phosphorylation. Geoscience Frontiers 8, 329-335. DOI: 10.1016/j.gsf.2016.06.008. *Simulation relevance: Reactive reduced-phosphorus source and pH-dependent phosphorylation scenarios.*

[24] Huber, C. & Wachtershauser, G. (1998). Peptides by activation of amino acids with CO on (Ni,Fe)S surfaces: implications for the origin of life. Science 281, 670-672. DOI: 10.1126/science.281.5377.670. *Simulation relevance: Mineral/CO-driven peptide formation under hot anaerobic aqueous conditions.*

[25] Weiss, M.C. et al. (2016). The physiology and habitat of the last universal common ancestor. Nature Microbiology 1, 16116. DOI: 10.1038/nmicrobiol.2016.116. *Simulation relevance: Phylogenomic boundary condition supporting an anaerobic H2/CO2/FeS-rich ancestral physiology.*

[26] Moody, E.R.R. et al. (2024). The nature of the last universal common ancestor and its impact on the early Earth system. Nature Ecology & Evolution 8, 1654-1666. DOI: 10.1038/s41559-024-02461-1. *Simulation relevance: Recent LUCA age/genome/ecosystem reconstruction; useful late-stage target with substantial uncertainty.*

[27] Barge, L.M. & Price, R.E. (2022). Diverse geochemical conditions for prebiotic chemistry in shallow-sea alkaline hydrothermal vents. Nature Geoscience 15, 976-981. DOI: 10.1038/s41561-022-01067-1. *Simulation relevance: Perspective on shallow alkaline vents that combine vent chemistry with cycling, freshwater and potentially wet-dry conditions.*

[28] Tutolo, B.M. (2026). Rethinking the origin of life at seafloor hydrothermal vents. PNAS 123, e2613108123. DOI: 10.1073/pnas.2613108123. *Simulation relevance: Critical 2026 reassessment: in-situ pH, sulfur scarcity, system geometry/longevity and hydrocarbon kinetics; must be represented as alternative parameterizations.*

[29] Mulkidjanian, A.Y. et al. (2012). Origin of first cells at terrestrial, anoxic geothermal fields. PNAS 109, E821-E830. DOI: 10.1073/pnas.1117774109. *Simulation relevance: Major terrestrial-geothermal alternative based partly on cellular ion composition and membrane constraints.*

[30] Sutherland, J.D. (2017). Opinion: Studies on the origin of life - the end of the beginning. Nature Reviews Chemistry 1, 0012. DOI: 10.1038/s41570-016-0012. *Simulation relevance: Systems-chemistry perspective emphasizing coordinated environmental chemistry and cyanosulfidic routes to building blocks.*

[31] Barge, L.M. et al. (2021). Experimental approaches for testing the hypothesis of the emergence of life at submarine alkaline vents. Life 11, 777. *Simulation relevance: Review of experimental reactor, chemical-garden, electrochemical and microfluidic methods; useful methods catalog.*

[32] Wang, W. et al. (2024). CO2 fixation to prebiotic intermediates over heterogeneous catalysts. Accounts of Chemical Research. DOI: 10.1021/acs.accounts.4c00151. *Simulation relevance: Catalysis-focused review useful for choosing mineral surfaces, mechanisms and kinetic experiments.*

[33] Martin, W., Baross, J., Kelley, D. & Russell, M.J. (2008). Hydrothermal vents and the origin of life. Nature Reviews Microbiology 6, 805-814. *Simulation relevance: Classic integrated review distinguishing black smokers from cooler serpentinization-driven alkaline systems.*

[34] Proskurowski, G. et al. (2008). Abiogenic hydrocarbon production at Lost City Hydrothermal Field. Science 319, 604-607. *Simulation relevance: Field evidence relevant to abiotic reduced carbon and methane/hydrocarbon provenance at Lost City.*

[35] Jackson, J.B. (2016). Natural pH gradients in hydrothermal alkali vents were unlikely to have played a role in the origin of life. Journal of Molecular Evolution 83, 1-11. DOI: 10.1007/s00239-016-9756-6. *Simulation relevance: Earlier mechanistic critique of whether natural vent pH gradients can perform useful chemical work.*

Appendix A. Mechanism-to-source provenance crosswalk

Mechanism / claim	Primary evidence	Supporting / counter evidence	Suggested model role
Serpentinization produces H2/alkaline fluids	[5] [8] [9]	Tutolo geology critique [28]	Boundary forcing / rock reactor
Long-lived hydrothermal forcing	[7]	[28]	Prior on forcing duration
Mineral pores / precipitates create interfaces	[1] [10]	[8]	Dynamic geometry and mineral state
pH gradient can drive CO2	[11] [12]	[28] [35]	Competing electrochemical

Mechanism / claim	Primary evidence	Supporting / counter evidence	Suggested model role
reduction			implementations
Minerals catalyze reduced carbon synthesis	[13] [14] [15]	[32]	Reaction kinetics and yields
Metabolism-like networks can form nonenzymatically	[16] [17]	[30]	Reaction graph / network emergence
Pores can concentrate dilute solutes	[18] [19]	geometry-dependent extrapolation	Transport validation target
Mixed amphiphiles form vesicles under vent-like conditions	[20]	environment-specific limits	Membrane state transitions
Environmental gradients can precede membrane energetics	[3] [21]	[35] [28]	Evolutionary energy module
FeS may supply fixed nitrogen	[22]	low yield; surface preparation sensitivity	N resource module
Reactive phosphorus can enter prebiotic chemistry	[23]	requires environmental transport/condition compatibility	P resource module
Hydrothermal mineral chemistry can form peptides	[24]	not sufficient for heredity	Polymer branch
LUCA compatible with anaerobic H2/CO2 metabolism	[25]	broader/newer reconstruction [26]	Late-stage validation constraint
Strict deep-sea setting may be insufficient	[27] [28] [29] [30]	classical theory [2] [4]	Scenario competition

Appendix B. Recommended implementation order

1. 1. Curate a provenance-first parameter registry and reproduce three benchmark experiments: Preiner 2020, Hudson 2020 and Baaske 2007.
2. 2. Build a reduced spatial vent interface with two fluids, temperature, pH/speciation, H2/CO2, dynamic precipitate and a small carbon-reduction network.
3. 3. Validate against Lost City ranges while separately parameterizing Hadean ocean chemistry; never calibrate ancient chemistry by silently reusing modern seawater.
4. 4. Add Muchowska/Moran protometabolic edges and measure network persistence under continuous flow and washout.
5. 5. Add pore remodeling and concentration mechanisms; then run ensembles over geometry and geochemical uncertainty.
6. 6. Only after the geochemistry-to-protometabolism model is predictive, add amphiphile compartments and selection.
7. 7. Keep heredity modular; compare multiple origins-of-information hypotheses rather than picking RNA by default.

8. 8. Run the full campaign as competing environmental scenarios: deep vent, shallow vent, terrestrial geothermal, and connected-niche models.

Bottom line The most defensible Vivarium program begins with a validated geochemical flow reactor and asks when persistent chemical organization appears. The scientific payoff is not merely showing that vent chemistry can make biomolecules; it is identifying the parameter regimes and transition sequence in which geology can become an evolving chemical ecology - and determining where the vent hypothesis fails.

Source-verification note: bibliographic metadata, DOI identifiers and the quantitative statements highlighted in the parameter table were checked against publisher, PubMed/PMC, or institutional records available through October 2026. Values from laboratory studies are labeled as experimental outcomes and should not be interpreted as direct reconstructions of Hadean environmental concentrations.